Renewable energy and access inequality

September 2026

Introduction

Energy inequality persists as a significant and uneven challenge within the global economy, influencing broader issues of economic development. As international communities and nations pursue the decarbonisation of energy systems, this report aims to determine whether transitioning to renewable energy sources reduces energy disparity or maintains existing inequalities. Decentralised renewable technologies can extend energy access to rural and remote populations that centralised fossil fuel systems cannot reach. However, factors such as national income level, stage of development, and policy frameworks can substantially influence this relationship.

This report investigates whether a higher share of renewable energy can be associated with a lower energy inequality within countries. The energy inequality is defined in three dimensions:

1. Energy Access: the share of the population able to access energy services
2. Energy Price: the affordability of energy
3. Energy Consumption: the distribution of energy usage across income groups within a country

When the three dimensions are considered, it is noticeable that Energy Access provides the clearest relationship between national renewable deployment and intra-country distribution, while providing the most consistent country data. This forms the core of our analysis, while price and consumption are further investigated as supporting dimensions. The data analysis and results are framed as association rather than causality given the observational nature of the data.

The primary data source considered is the World Bank World Development Indicators, which compiles figures from national statistical offices and international bodies including the IEA. The dataset covers over 135 countries over the period 2000 to 2021, giving 2970 country-year observations. One of the key variables is renewable energy consumption, as a share of total energy consumption (EG.FEC.RNEW.ZS). The gap between urban and rural electrification rates (EG.ELC.ACCS.UR.ZS - EG.ELC.ACCS.RU.ZS) is also used as a proxy, with the equivalent gap in access to clean fuels and technologies for cooking (EG.CFT.ACCS.UR.ZS - EG.CFT.ACCS.RU.ZS) also added.

Furthermore, the electric power consumption per capita (EG.USE.ELEC.KH.PC) is used as a measure of national consumption, alongside GDP per capita (NY.GDP.PCAP.CD), rural population share (SP.RUR.TOTL.ZS) and World Bank income group and region. Price in the data is not inherently discussed, which will be mentioned later under the limitations section. It is also worth noting that although the national renewable share is useful, it acts inherently as a proxy for the decentralised deployment that actually drives access equity.

The intended audience is energy policymakers and international development agencies, who are in the process of weighing where renewable investment can deliver the greatest equity gain.

The report is structured starting with data preparation, then exploratory analysis, and the concluding results from the data.

1 Data and Methodology

1.1 Data source

Our primary data source for this report was the World Bank World Development Indicators (WDI), accessed on the 21st September 2026 [1]. The WDI data bank compiles figures from national statistical offices and international bodies. We sourced the specific access indicators used from the World Bank's Global Electrification Database and SDG 7 tracking data. Both are based on nationally representative household surveys, with energy figures created by the International Energy Agency. We first exported the data as an Excel download [2] after querying the following indicators from the data bank:

Code	Variable	Role
EG.FEC.RNEW.ZS	Renewable energy consumption (% of final energy)	Independent
EG.ELC.ACCS.UR.ZS	Access to electricity, urban (%)	Electricity gap
EG.ELC.ACCS.RU.ZS	Access to electricity, rural (%)	Electricity gap
EG.CFT.ACCS.UR.ZS	Access to clean cooking fuels, urban (%)	Cooking gap
EG.CFT.ACCS.RU.ZS	Access to clean cooking fuels, rural (%)	Cooking gap
EG.USE.ELEC.KH.PC	Electric power consumption (kWh per capita)	Context
NY.GDP.PCAP.CD	GDP per capita (current US$)	Control
SP.RUR.TOTL.ZS	Rural population (% of total)	Control

Table 1: WDI indicators used in our analysis

Our initial export covered all 217 economies listed in the data bank over the full available range of 1990 to 2025, producing a workbook of 7,812 country-year rows across our eight indicators. Inspecting the raw file, we found that data coverage varied substantially between indicators, countries and time periods. Notably, rural population share was complete, with GDP per capita missing in only 5.3% of rows. Access indicators, however, were considerably sparser, with 14.3% of urban and 18.5% of rural electricity observations non-existent. Both clean cooking indicators were missing in 41.9% of rows. Furthermore, renewable energy consumption share, our independent variable, was missing in 13.6% of rows.

To begin, we noticed that missing values were recorded in the WDI export as `..` rather than blank values. We therefore replaced these using a replace-all Excel functionality before any numerical

work. Our initial inspection made evident that our usable window would be considerably narrower than the 36 years initially exported, something we expand upon below.

1.2 Choice of time period

We determined our time data window by data availability. Notably, access to clean cooking fuels data is not reported before 2000, and coverage of the electricity access gap rises sharply at the same point, from around 156 countries in 1999 to 192 in 2000. On the other side of the date spectrum, renewable energy consumption share is fully reported only up to 2021 (the 2022 release covers 71 countries and no data is available after this). We therefore restricted the analysis to years between 2000 and 2021.

1.3 Constructing the inequality measures

This report and our research question dive into inequality within countries, but we initially noted that renewable energy consumption share is reported at a national level only. We decided that our dependent variable must therefore carry the intra-country distribution in itself. We constructed two measures for this, each defined by the urban value minus the rural value and taken as a value only when both are non-blank entries. These are shown below:

1. Electricity access gap = EG.ELC.ACCS.UR.ZS − EG.ELC.ACCS.RU.ZS
2. Clean cooking access gap = EG.CFT.ACCS.UR.ZS − EG.CFT.ACCS.RU.ZS

We note that a larger gap indicates greater inequality in access between urban and rural populations. We also observed that electric power consumption per capita is a national aggregate with no internal country distribution. We therefore use this as context for overall consumption levels, rather than a measure of inequality itself.

1.4 Sample construction

To isolate our usable data, we applied three filters to the full set of countries/economies. First, we excluded 77 countries whose rural electricity access did not fall below 99% across the period defined. These were predominantly high-income economies with universal access in both rural and urban areas and whose gap is structurally zero. We selected the 99% figure arbitrarily, as countries reporting this in both urban and rural areas have effectively achieved universal access, and retaining them would add observations whose gap reflects measurement imprecision rather than genuine inequality. We additionally excluded three countries with fewer than 15 of the 22 years of renewable share data (Kosovo, South Sudan and American Samoa). We judged the 15-year threshold, approximately two-thirds of the 22-year window, to be the minimum required to observe a trajectory rather than a specific point in time. (Notably, this filter was not binding, as only three countries/economies fell below it. Thresholds between 12 and 17 years would have produced the same exclusions.)

We applied the 15-year threshold to renewable share, our independent variable. We noted that coverage of the dependent variables varies by country, and 11 retained countries fell below the 15-year boundary on the electricity access gap. Within our final sample, renewable share is observed for 2,963 of 2,970 country-years, the cooking gap for 2,882 and the electricity gap for 2,761. We see electric power consumption per capita as the sparsest series at 2,085 observations, which is a further reason to treat it as context rather than as a core variable in our analysis.

1.5 Data quality

We note several limitations of our data. The household surveys underlying the dataset are infrequent in many countries, meaning that values between survey years are effectively modelled, so short resulting movements may reflect estimation rather than actual change. We also observed that a small number of observations show negative gaps for our inequality measures, where rural access exceeds urban. It is therefore difficult to discern whether this is survey noise in countries close to energy parity, or genuine rural advantage over urban access.

2 Exploratory Data Analysis

In order to investigate the data, the exploration is structured in four stages: the distribution of the access gaps in absolute terms, their trajectory over time, their relationship with renewable energy share at a fixed point in time, and the relationship between changes in both over the full period. The summary statistics for the final sample are presented in the table below:

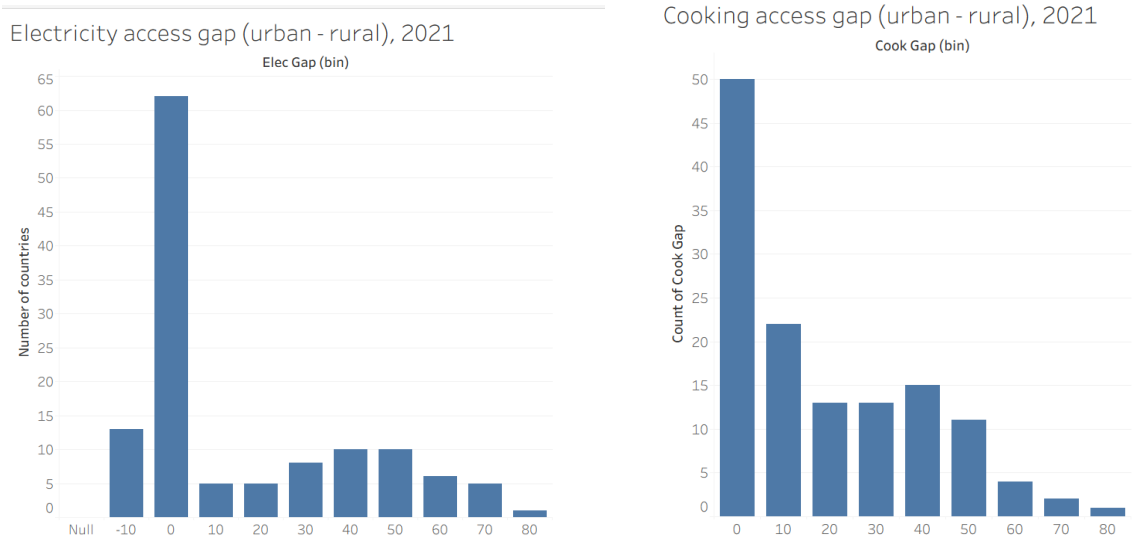
Variable	n	Mean	Median	SD	Min	Max
Renewable share (%)	2,963	40.0	33.7	30.8	0.0	98.3
Electricity gap (pp)	2,761	24.4	15.3	24.1	−8.2	99.2
Cooking gap (pp)	2,882	23.8	18.2	22.3	−33.3	86.4
GDP per capita (US$)	2,934	3,730	2,277	3,899	110	23,596
Rural population (%)	2,970	51.3	52.9	20.6	4.6	92.0
Electricity use (kWh per capita)	2,085	1,410	981	1,411	5	7,187

Table 2: Summary statistics

2.1 Distribution of access gaps

Figure 1 shows the reported distribution of both gaps at the fixed point in time chosen, 2021. It is noticeable that the electricity gap is polarised rather than evenly spread. 62 of 125 countries have a gap below 10 percentage points, 37 of them at exactly zero, while 40 countries retain a gap of 30 points or more. Therefore, it is noticed that the median gap of 2.0 is far below the mean value of 18.9. The 13 negative values are all between −6 and zero and are consistent with our earlier suggestion of survey noise, suspected in countries close to parity rather than genuine rural differences.

The cooking gap is more evenly distributed, with a median of 17.5 and 46 of 131 countries at 30 points or more. Therefore, the inequality in clean cooking is more widespread than inequality in electricity, a pattern that persists throughout the analysis.



(a) Electricity access gap

(b) Cooking access gap

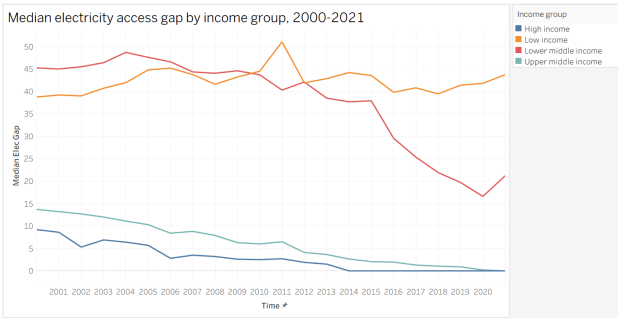
Figure 1: Distribution of the urban/rural access gap in 2021

2.2 Trends over time by income group

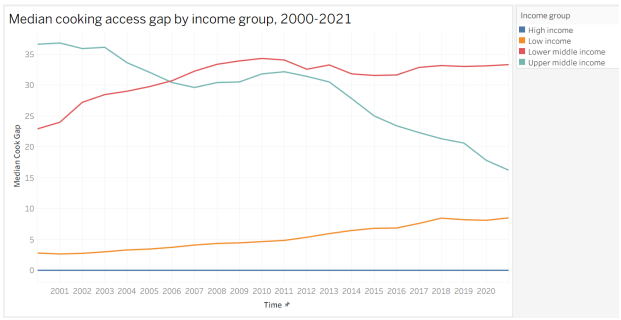
Figure 2 shows the median gap in each income group from 2000 to 2021. We see that for our electricity metric, high-income and upper middle-income countries converge on zero. Lower middle-income countries show little change until 2015 and then a sharp decline, from 38 to 21 points by 2021. The low-income median remains largely flat, moving from 40 to 44. We note that this series should be used with caution, as the number of low-income countries reporting both urban and rural access rises from 10 in 2000 to 22 in 2021. Therefore, we can possibly attribute some movement to changes in sample composition, rather than actual measured change within countries.

Interestingly, the cooking data revealed a limitation of our inequality measure that shaped the remainder of the analysis. The median cooking gap in low-income countries is the lowest of the three developing groups and rises over the period, from 2.8 to 8.5 points. This does not indicate equality, as access to clean cooking in these countries is close to zero in both urban and rural areas. We therefore hypothesise that the gap is small because almost no one has access in both groups. The subsequent rise in our data therefore may reflect urban households gaining access first. We then understood and concluded that a small gap for our metric can indicate either near-universal access or near-universal deprivation. This means the gap measure must be interpreted alongside the level of access to provide effective analytical conclusions.

By contrast, the upper middle-income median falls from 37 to 16 points, which does reflect rural catch-up, while the lower middle-income median remains around 33 points throughout.



(a) Median electricity gap



(b) Median cooking gap

Figure 2: Median urban/rural access gap by income group, 2000–2021

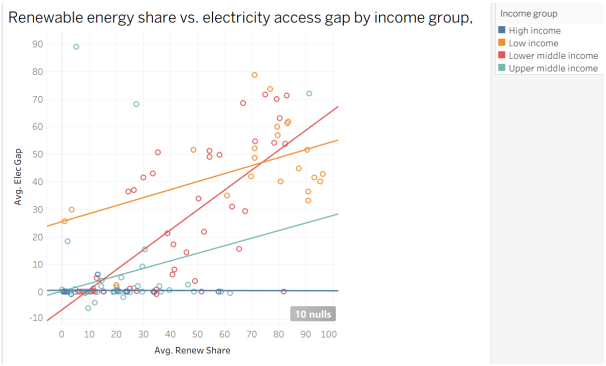
Over time, we therefore conclude that access gaps have closed where rural access has caught up, most clearly in richer groups. We also observe that in the poorest countries, low gaps reflect shared deprivation rather than equality.

2.3 Renewable share and the access gap

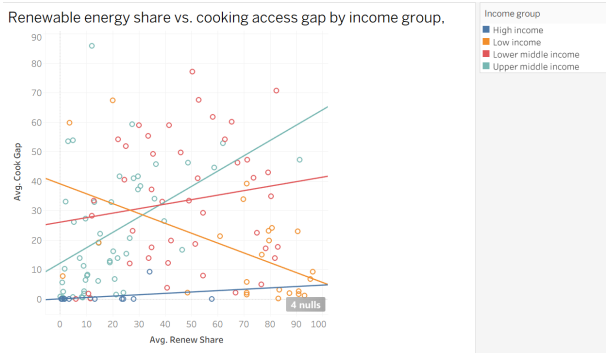
Figure 3 plots each country's renewable energy share against its electricity access gap in 2021, separately coloured by income group. Across all countries, we see that the relationship is strongly positive, with a Spearman correlation of 0.65. Countries with a higher share of renewable energy have larger urban/rural gaps. The same relationship for clean cooking is weaker but also notably positive at 0.28.

This is the opposite of what our initial hypothesis predicted. We believe this is explained by income: renewable share is strongly negatively correlated with GDP per capita (-0.66), and GDP per capita is itself negatively correlated with the electricity gap (-0.65). In other words, poorer countries tend to have both a higher renewable share and a larger access gap, which produces the positive relationship above even without renewables playing any direct role. In the poorest countries, a high share of renewables reflects reliance on traditional biomass such as wood and charcoal, which our chosen indicator counts as renewable, rather than modern sources such as solar or wind. When we compare countries within their own income group, we see that the relationship largely weakens.

Two outliers in our data demonstrate the limits of income group as a measure of development for this analysis. Equatorial Guinea has a renewable share of 5% and an electricity gap of 89, while Botswana has a share of 27% and a gap of 68. Both are classified as upper-middle income from our source; however, both still have large rural access deficits.



(a) Electricity access gap



(b) Clean cooking access gap

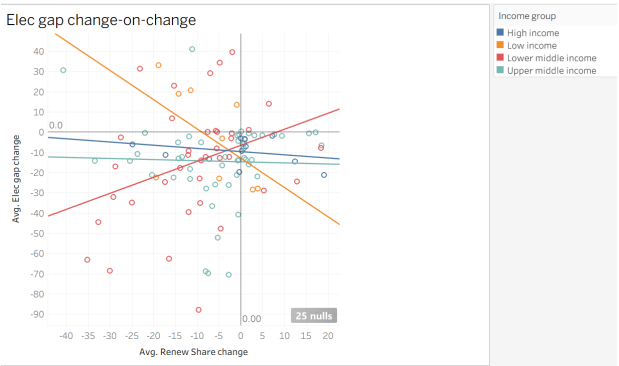
Figure 3: Renewable energy share against the urban/rural access gap by income group in 2021, with linear trend lines across each group

2.4 Change in renewable share and change in the access gap

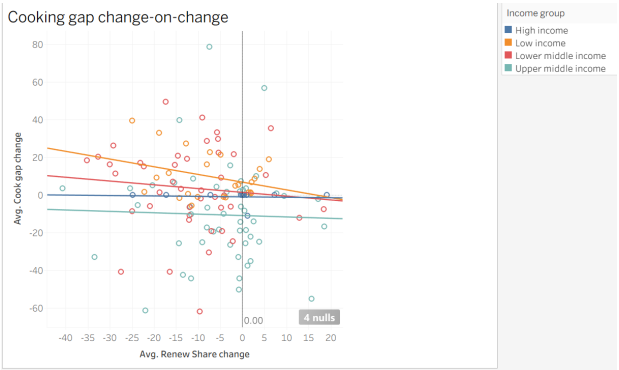
We note that comparing countries at a single point in time cannot separate the role of renewable energies from differences between countries. Figure 4 therefore compares how much each country's renewable share changed between 2000 and 2021 with how much its electricity gap changed over the same period.

The main pattern observed was that access became more equal while renewable share fell. Renewable share fell in 76 of the 108 countries, the electricity gap narrowed in 90, and 61 countries experienced both. Across all countries, there is no significant relationship between the two changes, with a p-value of 0.27.

Within our income groups, only lower-middle-income countries show a significant relationship (38 countries, $p = 0.039$, $R^2 = 0.11$). The slope for these was positive, meaning that the countries whose renewable share fell the most also saw their gap narrow the most. This fits our understanding of transition away from traditional biomass energy towards grid electricity, in which renewable share falls and rural access rises. We treat this result as tentative given it is the only significant relationship found out of all eight we tested.



(a) Electricity access gap



(b) Clean cooking access gap

Figure 4: Change in renewable share against change in the urban/rural access gap, 2000-2021, by income group

3 Results

Our findings do not support the hypothesis that a higher share of renewable energy is associated with lower energy inequality within countries.

First, within-country inequality in electricity access is polarised. By 2021, 62 of 125 countries had an urban–rural gap below 10 percentage points, while 40 retained a gap of 30 points or more. Gaps closed over the period in high-income and upper-middle-income countries and fell sharply among lower-middle-income countries after 2015, but remained around 40 points in low-income countries. Inequality in clean cooking access is more widespread and has narrowed more slowly.

Second, across all countries, renewable share is positively associated with the access gap. This means countries using more renewable energy have more unequal access. This relationship is largely explained by income, as within income groups, it weakens substantially and is significant in only two of eight cases.

Third, over the entire data period, access became more equal while renewable share generally fell. Of 108 countries with data at both endpoints, 76 saw their renewable share decline, and 90 saw their electricity gap narrow, with 61 experiencing both. The only significant relationship between these changes, among lower-middle-income countries, runs in the opposite direction to our hypothesis, where we see a larger fall in renewable share accompanied by a larger reduction in the gap.

	Level, 2021 (Spearman)		Change, 2000–2021 (p-value)	
Income group	Electricity	Cooking	Electricity	Cooking
All countries	0.65***	0.28***	0.27	0.14
High income	−0.07	0.39	0.37	0.80
Upper middle income	0.20	0.55***	0.83	0.82
Lower middle income	0.66***	0.16	0.039*	0.52
Low income	0.16	−0.40	0.14	0.15

Table 3: Association between renewable share and the urban/rural access gap. Level columns report Spearman correlations in 2021; change columns report p-values of linear trends between the change in renewable share and the change in each gap. * $p < 0.05$, *** $p < 0.001$.

4 Discussion and Limitations

Our findings are best explained by what the renewable share indicator measures. It counts traditional biomass such as wood and charcoal, which dominates energy use in low-income countries. As countries develop, households move from biomass to grid electricity, so renewable share falls while access improves, which is the pattern we observe. Our results therefore say little about modern renewables: decentralised solar may narrow rural gaps, but its effect cannot be separated from biomass in the national indicator.

The cooking data also showed that a small gap can reflect shared deprivation rather than equality. Median clean cooking access in low-income countries in 2021 was 13.0% urban and 0.5% rural. For policymakers, national renewable share is therefore a poor indicator of progress towards energy equity; rural access levels and the share of modern renewables are more informative.

Several limitations apply. Only the access dimension of our definition could be measured within countries: price is not reported in the WDI and consumption per capita has no internal distribution. Access data are survey-based, with values between surveys modelled, and coverage varies, with low-income electricity data rising from 10 to 22 countries over the period. Income classification reflects current status and was applied across the whole period. Methodologically, our analysis is observational, so we report associations rather than causal effects and cannot rule out confounding by factors such as conflict: Libya's rural electricity access, for example, collapsed from 99% to under 1% between 2000 and 2012 amid civil conflict, illustrating how country-specific shocks unrelated to renewable share can drive large movements in our outcome variable. The single significant change relationship came from eight tests and should be treated as tentative.

A fixed-effects panel regression would test whether the associations hold once stable country differences are controlled for, and is the natural next step.

5 Conclusion

This report asked whether a higher share of renewable energy is associated with lower energy inequality within countries, focusing on the gap between urban and rural access to electricity and clean cooking across 135 countries from 2000 to 2021. We find no evidence that it is. Across countries, higher renewable share is associated with larger access gaps, a relationship driven largely by income, and over two decades access became more equal in most countries even as their renewable share fell.

The main lesson is that renewable share, as currently measured, combines two very different things: traditional biomass, which is a marker of energy poverty, and modern renewable technology, which may be a route out of it. Treating them as one obscures any relationship between renewable energy and energy equity. Our analysis also shows that urban–rural gaps must be read alongside access levels, since the smallest gaps are sometimes found where almost no one has access.

For policymakers and development agencies, we recommend tracking rural access directly and measuring modern renewable deployment separately from traditional biomass, rather than relying on national renewable share as a measure of equitable progress. Future work should use data that isolate modern renewables, apply fixed-effects panel models to strengthen the evidence on causality, and draw on household survey microdata to cover the price and consumption dimensions of energy inequality that national indicators cannot capture.

References

[1] World Bank. (2026). *World Development Indicators*. Washington, DC: World Bank. `https://datatopics.worldbank.org/world-development-indicators/` (Accessed: 21 September 2026; data last updated 13 July 2026).

[2] World Bank. (2026). *World Bank Country and Lending Groups*. Washington, DC: World Bank. `https://datahelpdesk.worldbank.org/knowledgebase/articles/906519` (Accessed: 21 September 2026).